Expression substitution - mixed practice

Mix substitution into a single expression with collecting like terms.

Name: _____ Suggested time: 10 minutes

Show your working. Substitute carefully and simplify.

01 Given x=5y-5

Write in terms of y and simplify

Simplify: 5x+5

02 Given a=6b-6

Write in terms of b and simplify

Simplify: 4a+2b

03 Given x=3y+5

Write in terms of y and simplify

Simplify: 3x

04 Given a=5b+6

Write in terms of b and simplify

Simplify: 5a+4b

Answer sheet

Expression substitution - mixed practice | 4 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 $25y-20$

02 $26b-24$

03 $9y+15$

04 $29b+30$

Worked solutions

Expression substitution - mixed practice

01 Given x=5y-5

5x+5

Substitute x=5y-5.

5(5y-5) +5

25y -25 +5

25y-20

02 Given a=6b-6

4a+2b

Substitute a=6b-6.

4(6b-6) + 2b

24b -24 + 2b

(24+2)b -24 = 26b-24

03 Given x=3y+5

3x

Substitute x=3y+5.

3(3y+5)

9y +15

9y+15

04 Given a=5b+6

5a+4b

Substitute a=5b+6.

5(5b+6) + 4b

25b +30 + 4b

(25+4)b +30 = 29b+30